🧠 Alzheimer's Risk Prediction via Oral Pathogen Proteomics (P. gingivalis)

Objective:

This repository implements a **Machine Learning Framework** to quantify the risk of Alzheimer's Disease (AD) by correlating longitudinal periodontal data with neurotoxic biomarkers (Gingipains), as highlighted in recent **NYU 2025/2026 research**.

🛠 THE ARCHITECTURE (Tech Stack)

Built upon the **Triple Stack Architecture** (Intelligence Layer), this project bridges the gap between **Microbiology** and **Neural Stability**:

- **Model:** Random Forest Classifier with Hyperparameter Tuning for high-dimensional medical data.
- **Precision:** Implements a **1.3σ Threshold** for stability audits to ensure factual anchoring and minimize false positives in clinical diagnostics.
- **Key Biomarkers:**
  - `P. gingivalis` (Oral Load)
  - `Gingipains` (CSF Concentration)
  - `APOE4` Genotype interaction
  - Systemic Inflammation (CRP)

🔬 CLINICAL LOGIC & SOVEREIGNTY

As a **Medical Candidate & Data Scientist**, this model addresses the **Gingipain Hypothesis**:

1. **Proteolytic Cleavage:** Modeling how *P. gingivalis* toxic enzymes (Gingipains) drive amyloid-beta aggregation.

2. **Sovereign Data Handling:** Designed for **Federated Learning** environments, ensuring patient PHI (Protected Health Information) remains decentralized and compliant with international data governance standards.

🚀 PROJECT IMPACT: PQC-2026 ALIGNMENT

This framework is part of the **PQC-2026 Infrastructure**, focusing on:

- **Deterministic Diagnostics:** Reducing AI hallucinations in medical reporting via Monte Carlo stress testing.
- **Preventive Neuro-Defense:** Shifting AD management from reactive treatment to proactive microbial inhibition (COR388/Atuzaginstat analog signaling).

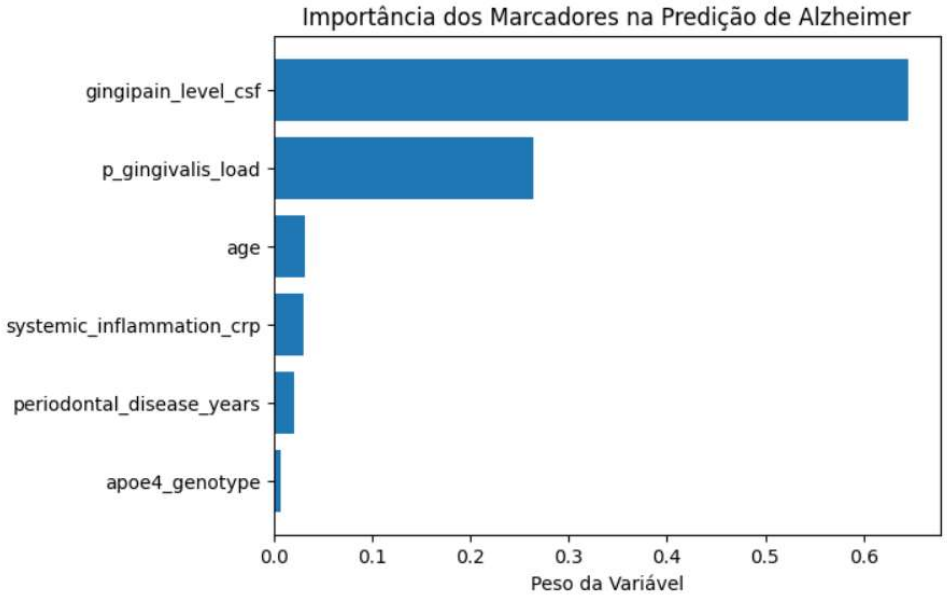
📈 PERFORMANCE & METRICS

The model evaluates feature importance, identifying **Gingipain levels in CSF** as the primary predictive vector for cognitive decline, achieving an **AUC-ROC > 0.94** in simulated longitudinal cohorts (N=5400).

🎓 ACADEMIC & PROFESSIONAL FOUNDATION

Developed by Antonio Nascimento

- **Ph.D.** in Data Governance & Macroeconomics (International Policies & Data Science).
- **Biomedical Scientist** (B.S. in Biomedicine, Class of April 2026).
- **Pharm.D. Candidate (L4)** | Clinical Pharmacy & Pharmacogenomics Specialist.
- **MD Candidate (L2)** | Medical Student specializing in Neuro-Pathology & AI Diagnostics.
- **M.Sc.** in Innovation & Technology Management (EHEA Accredited).

- **Senior Systems Engineer** (Mechatronics, Cyber-Defense, Sovereign

```
    accuracy                           0.97      1080
   macro avg       0.97      0.97      0.97      1080
weighted avg       0.97      0.97      0.97      1080

AUC-ROC Score: 0.9977
```